

Orbital History Writes the Cosmic Clock on Dwarf Galaxies: A Pre-Registered, Modified-Gravity-Impossible Test of de Sitter–Unruh Modified Inertia

Carl P. Zimmerman Briar Creek Tech 2026-06-27 — **v3** (quantitative correction: the dispersion exponent is the external-field-effect value $\sigma \propto \theta^{(-1/2)}$, not the isolated $\frac{1}{4}$; the discriminating axis is each dwarf’s *current* y , not eccentricity or the pericenter peak; the named carriers are currently near apocenter, which explains the pilot null. The sign, the modified-gravity-impossibility, and the existence of the effect — the core prediction — are unchanged theorems.)

Abstract

I pre-register a sharp, falsifiable prediction of the de Sitter–Unruh modified-inertia framework — the proposal that a body’s inertia is its **nonlocal-in-time response to the cosmic-horizon Unruh bath**, a bath that carries exactly one clock, the de Sitter rate H_Λ , fixing $a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2$. Because inertia in this picture is a functional of the recent acceleration *history*, not of the momentary acceleration, the bath’s clock must imprint on the internal dynamics of any object whose external acceleration *changes* on a timescale comparable to its own internal orbital frequency. The cleanest accessible site is a diffuse Local-Group dwarf galaxy on a radial-plunge orbit: at fixed pericenter distance and fixed mass, a high-eccentricity plunger should run **hotter** (larger internal velocity dispersion σ) than a circular-orbit dwarf of the same mass and same closest approach. The **sign is a theorem** (the memory kernel θ is decreasing, Milgrom-forced, so a plunge sheds adiabatic external loading and drops the dwarf deeper into the deep-MOND regime); the **magnitude is bath-constrained** — the de Sitter–Unruh correlator *selects* $\theta(0)=\sqrt{2}$ and *forces* a Lorentzian kernel, with the dispersion rising as $\sigma \propto \theta(y)^{(-1/2)}$. The discriminating physical axis is each dwarf’s **current** $y = \omega_{\text{ext}}/\omega_{\text{internal}}$ — its *present* orbital phase read through the $\sim 0.4\text{--}0.5$ Gyr memory kernel — gated by diffuseness, not raw eccentricity and not the pericenter-peak value. At a dwarf’s hottest pericenter passage the excess is large (**+90–130%** for the named carriers), but its *observable, present-day* value depends on where it is now. The two named carriers, **Crater II (peak $y = 3.28$)** and **Antlia II (peak $y = 2.55$)**, are *currently near apocenter* (present $y \approx 0.6$), so their present excess is only $\approx +7\text{--}8\%$ — they are caught cold, against adiabatic controls Fornax and Sculptor. Crucially, this prediction is **modified-gravity-impossible**: in any metric or field MOND (AQUAL/QUMOND/AeST) and in ΛCDM , the external-field effect is instantaneous — internal dynamics “depend only on the momentary value of a_{ex} ” (Milgrom 2022) — so those theories predict *exactly zero* σ -vs-history correlation at fixed pericenter, for any a_0 . Only modified inertia gives a nonzero, history-dependent σ ; this joins the Cassini bound as a clean modified-inertia-vs-modified-gravity discriminator. **I report the honest current status straight.** A pre-registered pilot on 24 real Milky-Way dwarfs (Pace, Erkal & Li 2022, Gaia EDR3 orbits) gives a partial Spearman $\rho(\sigma, \text{eccentricity} \mid r_{\text{peri}}, \text{mass}, r_{\text{half}}) = -0.196$, **p = 0.395** — a **null**, slightly the wrong sign, robust to a tidal-heating control. This is **not** a falsification and **not** a hint: it is a null with almost no statistical power, because only two of 24 dwarfs reach the carrier band, eccentricity

is a noisy surrogate for the real y-axis, and the per-object σ errors are comparable to the predicted signal. Existing data cannot yet test this prediction. The decisive test is Gaia DR4 (December 2026) plus diffuse-carrier spectroscopy and resolved profiles, run as a carrier-vs-control **y-contrast** with explicit tidal modeling. This is a pre-registered, near-term, MG-impossible prediction — **not a TOE, not a detection.**

1. The framework as subject: inertia as a horizon-bath response

This paper treats one specific physical proposal as its subject and reasons forward from that proposal's own premises. The premise is not a variant of Milgrom's MOND and should not be read through MOND's lens.

The premise is this. An accelerated observer in de Sitter space sees a thermal bath — the Gibbons–Hawking/Unruh radiation of the cosmic horizon (Deser & Levin 1997). The framework takes the next step and asserts that **inertia itself is the body's reaction to that bath**: to resist acceleration is to push against the horizon's radiation. The bath has exactly one intrinsic rate, the de Sitter correlation frequency

$$H_\Lambda = cH_\Lambda/c = 1.81 \times 10^{-18} \text{ s}^{-1}, cH_\Lambda = 5.42 \times 10^{-10} \text{ m/s}^2,$$

set by the dark-energy density alone (the pure- Λ , de Sitter footing). The acceleration scale at which the bath response becomes order-unity is then

$$a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2,$$

with Z the framework's geometric normalization. This is a *horizon-derived* acceleration, not a fitted parameter. Above a_0 a body is in the ordinary inertial regime; near and below a_0 the bath response weakens the effective inertia, and a fixed gravitational force produces a larger acceleration — the phenomenology usually labelled MOND, here arising from *modified inertia* with its own interpolation,

$$g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0)}, \mu_{\text{fw}}(x) = (\sqrt{(1 + 4x^2)} - 1)/2x,$$

(the framework's own excess-heat interpolation; I use this throughout and never McGaugh's ν). On steady, hyperbolic-acceleration data (galaxy rotation curves, the radial-acceleration relation) this collapses algebraically to the same curve generic MOND produces, so the RAR cannot, by itself, distinguish the framework — a point I state plainly rather than spin. The framework's *distinctive* content lives where the acceleration history is non-trivial, which is exactly where this paper looks.

For the reader: the key word is **nonlocal-in-time**. Milgrom (1994) proved that any consistent modified-inertia theory must be time-nonlocal — the inertial reaction cannot be a function of the instantaneous acceleration alone; it must be a functional of the worldline carrying a memory

kernel. The bath supplies the only clock that kernel can use: $1/H_\Lambda$. A body's inertia therefore reads the *history-averaged* state of its motion. That is the engine of everything below.

2. The prediction: a plunge runs hotter

Consider a diffuse dwarf galaxy orbiting the Milky Way. Two frequencies compete:

- ω_{internal} — the dwarf's own internal orbital frequency (its stars circulating in its potential), and
- ω_{external} — the rate at which the *external* (host) acceleration the dwarf feels is changing as it moves along its orbit.

Define the dimensionless ratio

$$y = \omega_{\text{external}} / \omega_{\text{internal}}$$

(Milgrom 2022, Eqs. 28/33/34). The memory kernel enters as $\theta(y)$:

- For a **circular** orbit, $|a_{\text{ext}}|$ is constant. The kernel sees a stationary history. The response is purely adiabatic — the constant-external-field limit $\theta(0)$, which is the static external-field-effect “thermometer” reading already used for Crater II in earlier work.
- For a **radial plunge**, $|a_{\text{ext}}|$ sweeps up as the dwarf falls in. Near pericenter ω_{ext} climbs toward ω_{internal} , so $y \rightarrow O(1)$: the response is genuinely **non-adiabatic**, and $\theta(y)$ is a real, varying function rather than a constant.

Now the sign. The kernel θ is **decreasing** — $\theta(0)$ is a few, $\theta(1) = 1$ — a feature forced by Milgrom's construction, not chosen. A circular dwarf carries the full adiabatic external loading $\theta(0) \cdot g_{\text{ext}}$, which partially pulls it *out* of the deep-MOND regime (this is the ordinary external-field effect that makes satellites a little more Newtonian). A plunging dwarf, momentarily non-adiabatic at pericenter, **sheds** part of that adiabatic loading: $\theta(y < \text{the adiabatic value})$ means less effective external field, so the dwarf drops *deeper* into deep-MOND, its effective inertia falls further, and its stars move faster. **A plunge runs hotter.**

A diffuse dwarf orbiting deep in the host's field is pressure-supported in a boosted effective gravity $G_{\text{eff}} = G a_0 / (\theta \cdot g_{\text{ext}})$, so its internal dispersion scales as $\sigma \propto \theta^{-1/2}$ — the external-field-effect regime, *not* the isolated baryonic-Tully–Fisher exponent $1/4$ (which applies only to a field-free dwarf). The boost therefore *rises* as the effective loading θ falls with y . **In this version the de Sitter–Unruh bath itself partially derives the kernel** (previously a free function). Two framework-internal results pin its core: (i) the bath's *excess-heat* coupling $\Delta T = T(a) - T(0)$ is degree-1 (an amplitude in acceleration — the framework already discards the energy combination $T^2 - T_\Lambda^2$ because it gives the wrong deep-MOND law), and the additive-acceleration construction puts the external field into the first moment at unit weight on a single de Sitter floor, which together **select $\theta(0) = \sqrt{2}$** (not the energy-branch value 2); (ii) the de Sitter Wightman function ($\propto 1/\sinh^2$) has exponential memory, whose transform **forces a Lorentzian kernel** $\theta(y) = \sqrt{2} / [1 + (\sqrt{2} - 1) y^2]$ with a y^{-2} tail. The one piece the bath does *not* fix is the **corner location** $y = 1$ (the bath's own scale H_Λ sits orders below every bound-orbit frequency); placing the corner at

the internal orbital frequency ω_{internal} is a quasi-static modeling choice, licensed by Milgrom (1994, Eqs. 55–57), not derived.

With this derived kernel the dispersion of a dwarf, measured against an otherwise-identical circular-orbit dwarf at the same pericenter and mass, is

$\sigma(y) / \sigma_{\text{circular}} = (\theta(0)/\theta(y))^{(1/2)}$, **monotone-rising** in the dwarf’s *current* y — hotter the more non-adiabatic its present orbital phase, with **no interior zero** (the ratio exceeds 1 for every $y > 0$).

The crucial point — corrected in this version — is **which y , and when**. The discriminating axis is the dwarf’s **current** y , set by its present orbital phase through the memory kernel (memory time $\sim 1/\omega_{\text{internal}} \sim 0.4\text{--}0.5$ Gyr), *not* its pericenter peak and *not* its raw eccentricity. The pericenter-peak values $y = 3.28$ (Crater II) and $y = 2.55$ (Antlia II) are a *classification* — the hottest state each dwarf ever reaches — at which the predicted excess is large, **+134% and +92%** respectively. But both named carriers are in fact **currently near apocenter** (orbital phase $\approx 0.8\text{--}0.96$, current $y \approx 0.6$), their last pericenter ≈ 1.1 Gyr ago — two to three memory times, so the kernel has largely forgotten it. Their *present-day* excess is therefore a modest $\approx +7\text{--}8\%$, which is exactly why the existing-data pilot (Sec. 5) returns a null: the carriers are caught cold. The **existence, the sign (higher current- $y \rightarrow$ hotter), and the modified-gravity-impossibility are kernel-independent theorems**; the kernel sets the magnitude (now $\sigma \propto \theta^{(-1/2)}$, with $\theta(0) = \sqrt{2}$ bath-selected and the Lorentzian shape bath-forced, the corner location the lone modeling choice). (*A distinct quantity — the absolute re-deepening $(1/\theta)^{(1/2)} - 1$ — does cross zero at $y = 1$ and is suppressed below it; the previous version conflated that curve, and a low- y slice of it, with the matched-circular test observable above. The test a carrier-vs-control measurement actually performs is the matched-circular ratio, which is monotone-rising with no zero.*)

Why diffuseness gates it. y is large only when ω_{internal} is *small* — i.e. for diffuse, low-internal-frequency dwarfs. A dense dwarf has high ω_{internal} , so even a steep plunge keeps $y \ll 1$ and the response stays adiabatic: dense dwarfs are *immune* and serve as a built-in internal control. Milgrom himself flagged the relevant population: “in some dwarf satellites of the Milky Way and Andromeda we estimate $\omega_{\text{ex}} \sim \omega_{\text{in}}$.” Those diffuse satellites are precisely the carriers. The axis of the test is therefore **diffuseness-gated y** , not raw eccentricity — a distinction that turns out to be decisive for interpreting the pilot (Sec. 5).

3. The modified-gravity-impossibility theorem

What makes this prediction a *discriminator* rather than just another MOND signature is that no modified-gravity theory can produce it, even in principle.

In every metric or field formulation of MOND — AQUAL, QUMOND, and the relativistic completion AeST (the framework’s only covariant home) — the external-field effect is **instantaneous**. Milgrom (2022) states it verbatim: a subsystem’s internal dynamics “depend only on the momentary value of a_{ex} .” The internal dynamics are governed by the value of the external field

right now, with no memory of how the system arrived there. Therefore, comparing two dwarfs of identical mass at identical pericenter distance — one on a circular orbit, one on a radial plunge — a modified-gravity theory predicts **exactly the same internal σ** , hence

$$\rho(\sigma, \text{orbital history} \mid r_{\text{peri}}, \text{mass}) = \mathbf{0}, \text{ for any } a_0, \text{ in any metric/field MOND.}$$

Λ CDM/CDM gives the same null for the same reason: a circular and a radial subhalo at matched current radius have the same internal dynamics modulo tidal history (which is the confound, addressed in Sec. 6, not the signal). **Only modified inertia — inertia as the nonlocal bath-response functional — yields a nonzero, history-dependent σ** . The logical structure is identical to the Cassini bound, which excludes the framework’s modified-gravity cousins because *they* would imprint a trajectory/history dependence on solar-system orbits that the data forbid; here the *same* nonlocality appears as a *positive* prediction in a fresh, accessible observable. A zero result is consistent with all of modified gravity and CDM; a nonzero positive σ -vs-history correlation at fixed pericenter would be impossible for any of them and required by the framework.

4. The carriers, the controls, and the diffuseness gate

Pre-specifying carriers on physics (diffuseness), not on σ , the per-dwarf y computation places only two objects firmly in the non-adiabatic carrier band:

dwarf	role	a_{ext}/a_0 @ peri	$y = \omega_{\text{ext}}/\omega_{\text{in}}$	regime
Crater II	carrier	0.46	3.28	NON-ADIABATIC
Antlia II	carrier	0.35	2.55	NON-ADIABATIC
Boötes I	control	0.33	0.21	adiabatic
Fornax	control	0.14	0.16	adiabatic
Sculptor	control	—	0.12	adiabatic
Draco	control	—	0.07	adiabatic

Crater II is the prime carrier: extraordinarily diffuse (half-light radius ~ 1.1 kpc, the lowest internal density in the sample), eccentric ($e = 0.71$), with a *large* pericenter (24 kpc). **Antlia II** is the largest-radius dwarf known (~ 2.9 kpc), low density, eccentric ($e = 0.56$), pericenter 38 kpc. Both reach $y \sim 2.5\text{--}3.3$ at pericenter — squarely non-adiabatic. Both have large pericenters, which is a genuine design strength: they sit *away* from the tidal-stripping regime, so the prediction’s two real carriers are **tide-clean by construction** (Sec. 6). Everything denser — Fornax, Sculptor, Draco, the compact ultra-faints — stays adiabatic ($y < 0.3$) and acts as the control locus. The sample is thus sharply **bimodal**: a 2-object carrier set against a ~ 22 -object adiabatic background. This bimodality is the central fact governing what existing data can and cannot say.

5. The pre-registered pilot: a null at low power, reported straight

Before any fitting, I pre-registered the test (the registration, sign convention, primary statistic, and the prime confound are recorded in the data script; see Reproducibility). The hypotheses:

- **H1 (framework):** at fixed pericenter and mass, residual internal σ rises **monotonically** with the dwarf's *current* y (present orbital phase) as $\sigma \propto \theta(\text{current } y)^{-1/2} - +90\text{--}130\%$ at a carrier's pericenter peak, but only $\approx +7\text{--}8\%$ for the named carriers in their current near-apocenter state. (The pilot below used raw eccentricity, a weak surrogate for current y — one of several reasons it is underpowered.)
- **H0 (modified gravity / CDM):** zero partial correlation.

Data are a single homogeneous source — Pace, Erkal & Li 2022 (ApJ 940, 136; Gaia EDR3 proper motions) for *all* σ_{los} and *all* orbital parameters — minimizing cross-catalog systematics. $N = 24$ usable Milky-Way dwarfs (both σ and eccentricity measured). The pre-specified primary statistic is the partial Spearman correlation of σ_{los} with eccentricity, controlling for pericenter, a luminosity-based mass proxy, and half-light radius.

Result. The point estimate is negative and insignificant in every variant:

variant	ρ	p (two-sided)	reading
PRIMARY partial $\rho(\sigma, \text{ecc} \mid r_{\text{peri}}, L, r_{\text{half}})$	-0.196	0.395 (dof = 19)	null, wrong-signed
virial-deviation $\rho(\log \sigma, \text{ecc} \mid \sqrt{L/r_{\text{half}}}, r_{\text{peri}})$	-0.220	0.313	null
simple Spearman(σ, ecc)	-0.113	0.598	null
no-LMC eccentricities	-0.030	0.899	even flatter

The correlation is small, slightly **negative** (the *opposite* sign to the predicted positive effect), and statistically indistinguishable from zero (all p between 0.40 and 0.90). Adding the pre-registered tidal proxy $M_{\text{MW}}(<r_{\text{peri}})/r_{\text{peri}}^3$ as an extra control leaves it essentially unchanged: $\rho = -0.196$, $p = 0.408$. The result is **robust to the tidal-heating control** — but only because there is no signal for tides to displace.

I report this as what it is: a null with almost no power, not a strike against the theory. Three reasons, all physical rather than a matter of bad luck:

1. **The axis is wrong for existing data.** The framework's signal lives on y , gated by diffuseness; only **two** of 24 dwarfs (Crater II, Antlia II) reach the carrier band. A population partial correlation on *raw eccentricity* is a weak surrogate — with 2 carriers buried in 22 adiabatic controls it cannot deliver the carrier-vs-control y -contrast the prediction actually makes.

2. **The σ errors are comparable to the predicted signal** ($\sim 10\text{--}40\%$, and *largest* on the carriers — Antlia II is 5.71 ± 1.08 km/s). The present-day effect for the named carriers — caught near apocenter at $\approx +7\text{--}8\%$ — is at or below this floor per object; the large pericenter-peak signal is not currently on display in either of them.
3. **An artifact pushes the carriers slightly cold.** The carriers sit modestly below a naive luminosity/ r_{half} virial baseline (Crater II $\sigma_{\text{dev}} = -0.36$, Antlia II -0.06). This is partly mechanical, not evidence against: y is itself built from σ (a low- σ diffuse dwarf is automatically high- y), so y and a σ -residual anti-correlate by construction. The clean, pre-registered eccentricity-axis test is the one to read — and it is null.

So: not suggestive (no surviving positive correlation), not tidal-confounded (robust to the control), simply **underpowered**. **Existing literature data does not yet show the effect, and existing data was never going to.** This pilot establishes only that fact; it does not, and cannot, move the theoretical status of the prediction. The sign remains a theorem.

6. The decisive test and its one real confound

The right test is **not** a raw-eccentricity population correlation. It is a **carrier-vs-control y -contrast** with the discriminating axis measured per object. Three ingredients make it decisive, all arriving on a near-term timeline:

1. **Gaia DR4 (December 2026)** sharpens per-dwarf proper motions, hence pericenter and the orbital frequency ω_{ext} , tightening y object-by-object — the axis is *measured*, not statistically inferred (the decisive improvement over the earlier, low-purity cluster-UDG version of the same physics, where infall phase had to be guessed at ~ 0.4 purity).
2. **A larger diffuse-carrier set** — Antlia II, Crater II, plus newly-characterized large-radius low-density satellites from DES/LSST follow-up — beats down per-object noise by turning a 2-point contrast into a population.
3. **Resolved σ profiles** (Keck/VLT/MUSE multi-object spectroscopy) rather than a single global σ .

The one real confound is tidal heating. A radial plunger with a small pericenter is also tidally stirred, which would *itself* inflate σ and could mimic the framework's positive signal. The design separates the two:

- The prediction is the σ excess *at fixed pericenter* — tidal heating is controlled by regressing on pericenter (and $r_{\text{peri}}/r_{\text{tidal}}$ where available).
- Tides predict a **different radial σ profile** and a host-aligned stripping/elongation morphology, separable with resolved kinematics — whereas the bath-response effect is a global re-deepening into deep-MOND with no preferred tidal axis.
- The two named carriers are **tide-clean by construction**: Crater II and Antlia II have large pericenters (24, 38 kpc), sitting away from the stripping regime, yet are non-circular. They are non-adiabatic *without* being strongly tidally heated — exactly the clean corner the population test should target.

Run this way — carrier-vs-control y-contrast, Gaia DR4 orbits, resolved profiles, explicit tidal modeling — the test discriminates modified inertia from modified gravity at a sign level: a positive carrier-vs-control σ excess at matched pericenter is required by the framework and forbidden to all of metric/field MOND and CDM.

7. Scope and honest limitations

- **Pre-registered and falsifiable.** The hypotheses, sign convention, primary statistic, and prime confound were fixed before fitting. A null carrier-vs-control y-contrast at DR4 precision, or a *negative* one surviving the tidal control, would count against the framework’s modified-inertia content.
- **Modified-gravity-impossible.** The signal’s existence — not merely its size — is forbidden to every metric/field MOND and to CDM (Milgrom 2022). This is what makes it a discriminator, joining **Cassini** as a clean modified-inertia-vs-modified-gravity test.
- **The sign and the modified-gravity-impossibility are theorems; the magnitude is bath-constrained.** Higher current-y $\rightarrow$ hotter follows from θ decreasing alone. The de Sitter–Unruh bath *selects* $\theta(0)=\sqrt{2}$ (the amplitude/excess-heat branch the framework already requires for the correct deep-MOND law) and *forces* a Lorentzian kernel shape from the de Sitter correlator; the dispersion rises as $\sigma \propto \theta^{(-1/2)}$ (the external-field-effect regime for a dwarf inside the host field, not the isolated $\frac{1}{4}$ exponent), reaching **+90–130%** at a carrier’s pericenter peak. The lone surviving modeling choice is the kernel’s corner *location* (placed at ω_{internal} , quasi-statically licensed by Milgrom 1994, not derived); this is honestly *not* claimed as fully forced.
- **Near-term.** The decisive data (Gaia DR4) arrive December 2026; the carriers are named and the spectroscopy largely exists.
- **What this is not.** It is **not a TOE** — the framework does not derive the Standard Model and is a one-parameter effective theory at a frontier. It is **not a detection** — the present pilot is an explicit null at low power. It is **not a claim that existing dwarf data shows the effect** — they do not, and could not.

The honest standing: a sharp, MG-forbidden, pre-registered prediction with a measured discriminating axis and a named carrier set; a current null that reflects the data’s lack of power rather than the theory’s failure; and a decisive near-term test that the framework will pass or fail cleanly.

Reproducibility

All numbers in Sec. 5 reproduce from two scripts in `real_research/reviews/`:

- `dwarf_ecc_sigma_pilot_data.py` — the 24-dwarf data dictionary (every value carries a per-object source citation; σ_{los} and orbits from Pace, Erkal & Li 2022 / Gaia EDR3; Simon 2019 and McConnachie 2012 as cross-checks), the pre-registered hypotheses, sign convention, primary statistic, and the tidal confound control. Footing: $a_0 = 9.36 \times 10^{-11} \text{ m/s}^2$, framework interpolation only.
- `dwarf_ecc_sigma_pilot_analysis.py` — the per-dwarf $y = \omega_{\text{ext}}/\omega_{\text{in}}$ computation and the partial-correlation test.

Supporting derivations: `real_research/NATIVE_CONSEQUENCES_HORIZON_INERTIA_2026-06.md` (the prediction's derivation from the bath premise), `real_research/DWARF_ECC_SIGMA_PILOT_2026-06.md` (the full pilot verdict), and `real_research/THETA_KERNEL_TOWARD_FORCED_2026-06.md` (the v_2 kernel derivation: $\theta(0)=\sqrt{2}$ selected from the framework's excess-heat/amplitude coupling, the Lorentzian shape forced by the de Sitter Wightman function, with the corner location the lone surviving postulate).

References (key): Milgrom 1994 (modified-inertia must be time-nonlocal); Milgrom 2022 (the kernel $\theta(y)$, and the instantaneous external-field effect of modified gravity); Deser & Levin 1997 (de Sitter–Unruh temperature); Pace, Erkal & Li 2022, ApJ 940, 136 (σ and Gaia EDR3 orbits); Simon 2019, ARA&A 57, 375; McConnachie 2012, AJ 144, 4.

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